

Robust Machine Learning versus Statistical Models for Commodity and Macroeconomic Risk Signals: Evidence from Ghana

Jonathan Nelson

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Abstract

Whether machine learning genuinely improves on classical econometric models for financial risk prediction, or merely appears to under validation protocols too permissive to detect overfitting, remains contested even in well-studied developed markets. We build an eleven-model comparison – four statistical models (a naïve base rate, an AR-GJR-GARCH- t exceedance-probability model, a vector autoregression, and a two-state Markov-switching model) against seven machine-learning classifiers (logistic regression, LASSO, elastic net, random forest, XGBoost, a support vector machine, and a small neural network held deliberately as a weak benchmark) – for three daily risk signals in a commodity-dependent emerging economy: cocoa value-at-risk breaches, cedi depreciation episodes, and high-volatility days. Using nested time-series cross-validation with nested hyperparameter tuning on a nine-year daily panel (2015–2026) spanning four commodity futures, the official Bank of Ghana exchange rate, global risk indicators, and Ghana macroeconomic series – including the real 2024 cocoa supply shock as an out-of-sample stress test – we find that every sophisticated model – statistical or machine learning – clearly and significantly outperforms naïve baselines, but that after correcting for the thirty-one hypothesis tests this comparison entails, no single model significantly outperforms the other serious contenders for any of the three targets. One partial exception survives full correction: random forest significantly beats the weakest baseline models for cedi depreciation, though not the other serious competitors. Permutation-importance and SHAP diagnostics corroborate this picture from an independent angle: feature importance for cocoa risk is statistically indistinguishable from noise across cross-validation folds, while cedi depreciation shows a modestly stable, economically interpretable pattern dominated by the currency’s own short-term momentum. We situate these findings against a closely parallel recent result for a G7 currency pair, and argue that the appropriate conclusion is not "machine learning wins" or "statistics wins," but that model risk – genuine uncertainty about which sophisticated model is correct – is itself the empirically supported finding, mirroring an analogous model-risk result in a companion study of the same economy’s commodity tail-dependence structure.

Keywords: machine learning, statistical forecasting, model risk, commodity risk, exchange-rate risk, nested cross-validation, Ghana, emerging markets

1. Introduction

Two roughly opposite failure modes recur across the literature comparing machine learning to classical statistical models for financial risk prediction. The first is methodological optimism: reporting a machine-learning model’s point-estimate advantage over a statistical baseline without testing whether that advantage is statistically distinguishable from noise, a gap the financial-machine-learning methodology literature has identified as pervasive (López de Prado, 2018; Vabalas et al., 2019). The second is insufficient statistical competition: comparing machine learning against an unregularized logistic regression or a plain autoregression, rather than against the genuinely best available classical alternative – a comparison that can make machine learning look stronger than it is simply because its statistical competitor was weak, not because the underlying task favors flexibility over structure.

This paper is designed to avoid both. On the first, every comparison in this paper is tested for significance by a paired, block-bootstrap procedure on pooled out-of-fold predictions, corrected for the full family of thirty-one hypothesis tests the comparison entails by both a family-wise Bonferroni correction and an independent Benjamini–Hochberg false-discovery-rate procedure – the same dual-correction standard applied in a companion study of this economy’s

commodity tail-dependence structure (Nelson, 2026), chosen deliberately so that a result surviving only one correction is treated as weaker evidence than a result neither touches, and so that the two papers’ inferential standards are directly comparable. On the second, the statistical side of this comparison is not limited to a single baseline: alongside an AR-GJR-GARCH- t exceedance-probability model, we build a vector autoregression – a genuine multivariate linear competitor capable of exploiting cross-market information a univariate GARCH structurally cannot – and a two-state Markov-switching model, reusing validated regime-switching machinery from the companion tail-dependence study’s treatment of the same currency’s difficult managed-float marginal distribution.

The setting is Ghana, chosen for the same reason as the companion study: cocoa, gold and crude oil jointly account for the large majority of the country’s export receipts, and the Bank of Ghana’s official interbank exchange rate – not the commonly used but empirically unreliable Yahoo Finance proxy, whose daily returns correlate with the official rate at only 0.18 – provides a genuine, policy-relevant currency risk signal. We test three daily binary risk events: a cocoa value-at-risk breach (next-day return below its trailing 5% quantile), a cedi depreciation episode (cumulative five-day depreciation above its trailing 90% quantile), and a high-volatility day (next-day absolute cocoa return above its trailing 95% quantile). All three are tested with an expanding-window evaluation design whose final holdout is the real 2024 cocoa supply shock – not a synthetic stand-in – providing a genuine out-of-distribution stress test rather than only an average-case cross-validated one.

Our design is deliberately conservative in a second sense: hyperparameters for every tunable model are selected by nested cross-validation, an inner expanding-window search confined strictly to each outer fold’s training data, following the methodology literature’s own diagnosis that standard k -fold cross-validation produces optimistically biased performance estimates in exactly the small, serially dependent samples financial applications typically provide (Vabalas et al., 2019). Where an inner split would contain too few positive examples to produce a reliable tuning signal – a concrete, measured risk we verify directly rather than assume away – that specific outer fold falls back to a documented default configuration rather than tuning on a signal known in advance to be unstable.

Our central contribution is not a claim that machine learning wins or loses this comparison, but a rigorously supported account of which claims the data can and cannot support. Every serious model, statistical or machine-learning, clearly and significantly outperforms naïve baselines. After correcting for multiple comparisons, however, no model significantly outperforms the other serious contenders for cocoa risk or high-volatility days, and only a partial result – random forest beating the weakest baselines, not the other serious competitors – survives for cedi depreciation. Permutation-importance and SHAP diagnostics corroborate this from an independent methodological angle: feature importance for cocoa risk is statistically indistinguishable from noise across cross-validation folds. We situate this against a closely parallel, contemporaneous finding for the CAD/USD exchange rate (caudsdstudy2026), where an almost identical expanding-window, SHAP-interpreted design found that ensemble machine-learning models fail to significantly beat a random-walk-adjacent statistical benchmark under the Diebold–Mariano test – evidence that our result is not an artifact of this paper’s specific emerging-market setting.

The remainder of the paper proceeds as follows. Section 2 reviews the machine-learning-versus-statistics literature for financial risk prediction, the currency-crisis early-warning-system literature specifically, and existing work on the Ghana cedi. Section 3 documents data construction. Section 4 sets out the model roster, nested cross-validation design, and significance-testing procedure. Section 5 reports results. Section 6 discusses implications. Section 7 states limitations. Section 8 concludes.

2. Related literature

2.1. Machine learning versus statistical models: a mixed and methodologically uneven literature

The broader literature comparing machine learning to classical econometric models for financial forecasting is large, fast-growing, and genuinely mixed in its conclusions – a heterogeneity that is itself informative. For continuous volatility forecasting, a substantial recent literature reports gradient-boosting and deep-learning models outperforming GARCH-family models on error metrics such as RMSE and MAE across equities, energy commodities and cryptocurrencies (Yaro et al., 2025; Yaro, 2024; Munich Personal RePEc Archive, 2025), with deep learning showing particular advantages at medium-to-long forecast horizons (structuralbreaks2025). A recurring methodological pattern in this literature, however, is that point-estimate error-metric comparisons are reported without testing whether the reported advantage is statistically distinguishable from the alternative model’s forecast error – a gap Hassanniakalager et al.

(2024) address directly, building a false-discovery-rate-based framework specifically to separate genuine "buckets" of superior volatility models from what could be sampling variation, a design instinct this paper's own dual-correction procedure shares.

2.2. *Currency-crisis and exchange-rate classification: closer to the present task, and more cautious in its conclusions*

Because this paper's task is binary tail-event classification rather than continuous forecasting, the currency-crisis early-warning-system (EWS) literature is a closer methodological comparison than the volatility-forecasting literature above. Results here are more mixed, and more favorable to classical and simple models than the volatility literature might suggest. Random forests and extremely randomized trees have been shown to outperform logistic regression for financial-crisis classification across multiple data sources and specifications (crisisrf2024), and random-forest-based approaches combined with wavelet transforms have been applied specifically to currency-crisis forecasting (rfcrisis2025). Against this, a closely relevant 2026 study using a thirty-country emerging-market panel (1998–2023) with LSTM and attention mechanisms found that *logistic regression and random forest outperform the LSTM model on AUC* (lstmattention2026) – direct precedent, on a comparable emerging-market panel, for this paper's own decision to retain neural networks as a deliberately weak benchmark rather than a central contender. An IMF working paper on fiscal-crisis prediction further documents that unregularized logistic regression specifically underperforms regularized and tree-based alternatives in smaller samples, since it has no built-in mechanism to address overfitting (International Monetary Fund, 2021) – a specific, citable explanation for this paper's own finding that unregularized logistic regression is the weakest model in the comparison for two of three targets, while its regularized counterparts (LASSO, elastic net) perform substantially better.

The most methodologically direct precedent for this paper's design and conclusion, however, is not in the emerging-market or currency-crisis literature at all. A contemporaneous study of the CAD/USD exchange rate (causdstudy2026) uses expanding-window evaluation and SHAP interpretability – the same two design choices this paper makes independently – together with the Diebold–Mariano test of equal predictive accuracy, and finds that every ensemble machine-learning model tested (random forest, gradient boosting, XGBoost, AdaBoost) fails to significantly outperform a random-walk-adjacent statistical benchmark, with only plain linear regression achieving modest, significant outperformance. That result, obtained independently on a liquid, data-rich G7 currency pair using near-identical methodology, is the closest available external check on whether this paper's own central finding is a genuine feature of the model-comparison problem or an artifact of this paper's specific emerging-market setting; the close correspondence between the two results is evidence for the former.

2.3. *Markov-switching models for exchange rates: a literature that has itself long been cautious*

The Markov-switching exchange-rate literature provides useful context for interpreting this paper's own Markov-switching results. Engel's foundational study (Engel, 1994) found that a Markov-switching model fit to eighteen exchange rates does not generate superior forecasts to a random walk by mean-squared error, though showed some promise specifically for directional prediction. More recent work examining whether additional regimes improve forecast accuracy finds that the regime count minimizing forecast error typically corresponds to the most parsimonious well-specified model, not the most flexible one (regimecount2019). This history of measured, rather than triumphant, claims for Markov-switching models in exchange-rate forecasting is consistent with, not contradicted by, this paper's own finding: Markov-switching achieves strong point estimates for two of three targets but does not significantly clear the bar against other serious competitors once corrected for multiple comparisons.

2.4. *The Ghana cedi specifically: existing work, and the gap this paper fills*

Machine learning applied to the Ghana cedi specifically is not new. Existing work includes an XGBoost model forecasting the GHS/USD, GHS/GBP and GHS/EUR exchange-rate *levels* (ghanaxgboost2025), an LSTM network forecasting weekly cedi rates using Google Trends search data alongside macroeconomic and commodity-price inputs (ghanalstm2021), and a Bayesian model averaging study of exchange-rate drivers that explicitly identifies commodity price fluctuations (gold, cocoa, crude oil) as a channel of Ghanaian currency exposure (ghanabma2024), directly corroborating this paper's and the companion tail-dependence study's shared premise. None of this existing work, to our knowledge, poses the comparison this paper poses: binary tail-event classification (rather than continuous level

forecasting), a genuine statistical-versus-machine-learning horse race spanning both paradigms with comparably serious competitors on each side, nested time-series cross-validation, and formal multiple-testing-corrected significance testing. That specific combination, rather than the application of machine learning to the cedi in general, is the gap this paper addresses.

2.5. *Feature-importance instability: a documented property of small-sample financial machine learning, not an anomaly*

The methodology literature on financial machine learning specifically warns that standard cross-validation is prone to optimistic bias in the small, serially dependent samples typical of financial applications (López de Prado, 2018; Vabalas et al., 2019), and separately documents that feature-selection stability and predictive accuracy are largely independent properties of a model: comparing eight feature-selection methods across multiple breast-cancer expression datasets, Haurly et al. (2011) find that the choice of selection method affects the *stability* of the resulting feature signature far more than it affects predictive accuracy. This paper’s own finding – near-zero cross-fold stability of permutation importance for cocoa risk, despite models that are not themselves broken or degenerate – is consistent with, rather than contradicted by, this documented property of small-sample, low-signal-to-noise prediction problems, and is a further reason to distinguish, as this paper does throughout, between a model’s predictive accuracy and the reproducibility of its internal explanation.

3. Data

The daily panel spans 5 January 2015 to 15 July 2026 and comprises seven series: ICE cocoa, COMEX gold, Brent and WTI crude futures (Yahoo Finance), the Bank of Ghana interbank USD/GHS mid-rate – the same official series validated and defended at length in the companion tail-dependence study (Nelson, 2026), in preference to the unreliable Yahoo GHS=X proxy – the CBOE VIX index, and the Federal Reserve’s Nominal Broad U.S. Dollar Index (DTWEXBGS), both sourced from FRED with full history (VIX since 1990, the dollar index since 2006) after an initial download inadvertently captured only a five-year default window, caught and corrected before use. Three monthly Ghana macroeconomic series are merged with a one-month publication lag: headline CPI inflation, merchandise export receipts (f.o.b.), and the Bank of Ghana Monetary Policy Rate. The policy-rate series required the same data-integrity correction already documented for exports in the companion study: the source table encodes an eight-month reporting gap (May–December 2023) as a literal value of 0.00 rather than a missing-data marker, which we detected, isolated to exactly those eight consecutive months, and masked to missing before use, following the identical procedure and reasoning applied to the analogous exports-table gap in the companion paper.

A separate data-integrity issue specific to this paper’s feature construction is documented here since it materially affects the analysis: an initial implementation’s macro-feature merge silently truncated the entire post-June-2023 sample – including the whole 2024 stress-test window this paper’s design depends on – because the source macro table contains explicit, dated rows with missing values extending to the present for the series that stopped updating, rather than simply ending; a naïve forward-fill-on-reindex operation propagated those missing values forward indefinitely rather than carrying the last known reading forward from before the gap began. The fix forward-fills each macro column’s own gaps prior to reindexing onto the daily calendar, restoring full coverage including all 262 trading days of 2024.

Three binary daily risk labels are constructed, each defined by a rolling, information-available-at- t threshold to avoid look-ahead contamination: a cocoa value-at-risk breach (next-day cocoa return below its trailing 250-day 5% quantile; event rate 5.4%), a cedi depreciation episode (cumulative depreciation over the next five days above its trailing 250-day 90th-percentile threshold, computed in the currency’s natural GHS-per-USD orientation and independently of the sign convention used elsewhere; event rate 12.5%), and a high-volatility day (next-day absolute cocoa return above its trailing 95th-percentile threshold; event rate 6.4%). The feature set comprises 95 leakage-free predictors per observation: lagged returns, rolling volatility and momentum at multiple horizons, cross-market rolling correlations, a joint-tail-count indicator, and the lagged macroeconomic block, computed identically for all seven daily series (Appendix Appendix A lists the full feature construction).

4. Methodology

4.1. Model roster

Four statistical models and seven machine-learning classifiers are compared, chosen specifically so that the statistical side is not limited to a weak competitor.

Base rate. Predicts the training-sample event frequency; the floor every other model must clear.

GARCH- t . An AR(1)-GJR-GARCH(1,1) model with Student- t innovations fit to the target series' own return history, converting the model's one-step-ahead conditional distribution into an exceedance probability against the same rolling threshold used to define the label. This is the genuine risk-desk benchmark a competent institution would deploy, and is structurally univariate: it cannot see cross-market information by construction.

Vector autoregression (VAR). Fit to the five core daily return series (cocoa, gold, Brent, WTI, cedi), deliberately excluding VIX and the dollar index to keep parameter count estimable in the smallest early cross-validation folds (parameters scale as k^2p); lag order selected by AIC, capped at 5, and empirically verified to adapt sensibly to fold size (selecting $p = 2$ on the smallest training fold against $p = 5$ on a larger one, rather than mechanically maximizing the cap regardless of context). The one-step-ahead forecast error covariance at horizon 1 is exactly the fitted innovation covariance matrix, giving a closed-form marginal forecast distribution for the target series without simulation. This is the multivariate linear statistical competitor GARCH cannot be by construction, motivated directly by this paper's own finding (Section 5.4) that cross-market information matters for cedi risk.

Markov-switching. A two-state Gaussian hidden Markov model fit to the target series, using one-step-ahead *predictive* regime probabilities – computed via the forward algorithm from information through $t - 1$ only – to weight the mixture distribution used for the exceedance probability, avoiding the look-ahead contamination that would result from using the observation itself to weight the distribution it is being evaluated against. This construction directly reuses the regime-switching machinery validated in the companion study's treatment of the cedi's own difficult marginal distribution (Nelson, 2026), here repurposed from a diagnostic tool to a forecasting model. The number of states is fixed at two rather than tuned, a deliberate, disclosed scope choice consistent with the regime structure already established for this currency in the companion study.

Logistic, LASSO, elastic net. Standard, LASSO (ℓ_1), and elastic net (ℓ_1/ℓ_2 , mixing ratio 0.5) logistic regression, each on standardized features; separated into three models rather than one, since regularization type is itself a variable of interest given this paper's finding (Section 5) that unregularized logistic regression is the weakest model in the entire comparison for two of three targets.

Random forest, XGBoost. Standard ensemble tree methods. XGBoost's `min_child_weight` parameter required a specific, documented correction during development: an initial value calibrated against a raw sample-count intuition (appropriate for scikit-learn's `min_samples_leaf`, a different quantity) was far too restrictive against the roughly 55 positive training examples available in the smallest cross-validation fold, collapsing every tree to a single leaf and every prediction to the training base rate – caught because the resulting predictions were suspiciously degenerate (a single repeated value across a heterogeneous test set), not because the resulting AUC alone looked implausible.

Support vector machine. An RBF-kernel SVM with Platt-scaled probability outputs, on standardized features.

Small neural network. A single-hidden-layer multilayer perceptron (8 units), included specifically as a weak benchmark rather than a central contender, consistent with a deliberate scope decision made before this paper's design was finalized. A larger, sequence-based architecture (e.g., an LSTM or GRU operating on raw return sequences rather than hand-engineered features) was considered and explicitly not built: a single GRU layer at modest hidden size already requires on the order of 3,600 trainable parameters, against a standard rule-of-thumb requirement of 10–50 training examples per parameter for stable training – 36,000 or more sequences – while this paper's entire daily sample is under 3,000 observations, with as few as roughly 150 positive examples for the rarest target across the full eleven-year sample. Published sequence-model successes in finance typically rely on cross-sectional pooling across hundreds of related series (e.g. Fischer and Krauss, 2018, pooling across the S&P 500's constituent stocks) to reach an adequate effective sample size, an option unavailable for a single emerging-market currency studied in isolation.

4.2. Nested time-series cross-validation

Five expanding-window outer folds plus a sixth, final holdout on calendar-year 2024 – the real cocoa supply shock, not a synthetic stand-in – provide the outer evaluation layer. For every tunable model, hyperparameters are selected by an inner expanding-window search (three inner folds) confined strictly to the outer fold's training data,

with the winning configuration refit on the complete outer training set before scoring on the untouched outer test set – the standard nested design specifically to prevent the tuning process itself from leaking information into evaluation (Vabalas et al., 2019).

A concrete risk in this design, verified directly rather than assumed away: the smallest outer training fold’s inner splits can produce as few as two positive test examples for the rarest target, a sample too small to produce a meaningful hyperparameter-selection signal. Any outer fold whose inner split would fall below ten positive examples in any inner test fold skips nested tuning for that fold entirely and uses a fixed, pre-specified default configuration instead, logged explicitly rather than silently substituted. Across all eighteen (fold, target) combinations and seven tunable models, this fallback triggered for 22% of tunable-model results, concentrated entirely in the earliest one to two outer folds and never in the final five-fold-average or the 2024 stress holdout – confirming the fallback protected exactly the folds it was designed to protect without contaminating the folds that carry this paper’s conclusions.

4.3. Significance testing

Point-estimate AUC comparisons are tested for significance by a paired moving-block bootstrap (block length 20 trading days, 2,000 replicates) on pooled out-of-fold predictions: because every model in a given comparison is scored against the same realized outcomes on the same dates, a paired test exploits this shared structure rather than treating the two AUC estimates as independent. For each of the three targets, the model with the highest *pooled* AUC – computed over all out-of-fold predictions concatenated across outer folds, the standard and more conservative choice relative to averaging per-fold AUCs separately, since it evaluates one ranking over the complete held-out sample rather than averaging metrics of very different individual variance – is tested against every other model, plus explicitly against GARCH- t regardless of rank, as the natural "does anything beat the standard statistical baseline" comparison. This produces thirty-one tests across the three targets, corrected by both a family-wise Bonferroni procedure and an independent Benjamini–Hochberg false-discovery-rate procedure, exactly mirroring the dual-correction standard of the companion tail-dependence study.

5. Results

5.1. Main findings

Three results carry this paper’s argument. First, **every serious model, statistical or machine-learning, clearly outperforms naïve baselines** across all three targets. Second, **after correcting for the full family of thirty-one tests this comparison entails, no model significantly outperforms the other serious contenders for cocoa value-at-risk breaches or high-volatility days**; the point-estimate leaders (Markov-switching and the vector autoregression, by pooled AUC, for the fold-averaged view; GARCH- t , by the more conservative pooled- AUC view) do not clear a rigorous bar against genuine competitors. Third, **for cedi depreciation, random forest’s advantage over the weakest baseline models survives full Bonferroni correction** – the most robust single result in this paper – **but its advantage over the other serious competitors (XGBoost, GARCH- t , the vector autoregression, Markov-switching) does not**. Permutation-importance diagnostics corroborate the cocoa result from an independent angle: cross-fold feature-importance stability for cocoa value-at-risk breaches is statistically indistinguishable from zero ($\rho = -0.009$), meaning the elastic-net model’s apparently informative coefficients do not reproduce across time windows and should not be read as a stable economic relationship.

5.2. Fold-averaged and pooled rankings, and why they differ

Table 1 reports mean AUC across the five expanding- window folds plus the 2024 stress holdout, for the three highest-ranked models per target. Table 2 reports the pooled-AUC ranking – the statistic the significance tests in Section 4.3 are built on – which differs from the fold-averaged view for two of three targets.

Fold-averaging weights every fold equally regardless of size or composition; pooling evaluates one ranking over the complete held-out sample and is the more standard, and in our view more defensible, summary statistic precisely because early folds’ AUC estimates carry much higher sampling variance (as few as several dozen positive examples) than later folds’, a discrepancy fold-averaging does not correct for. We report both for transparency, treat the pooled ranking as authoritative, and note explicitly where they disagree rather than presenting only the version that tells a cleaner story.

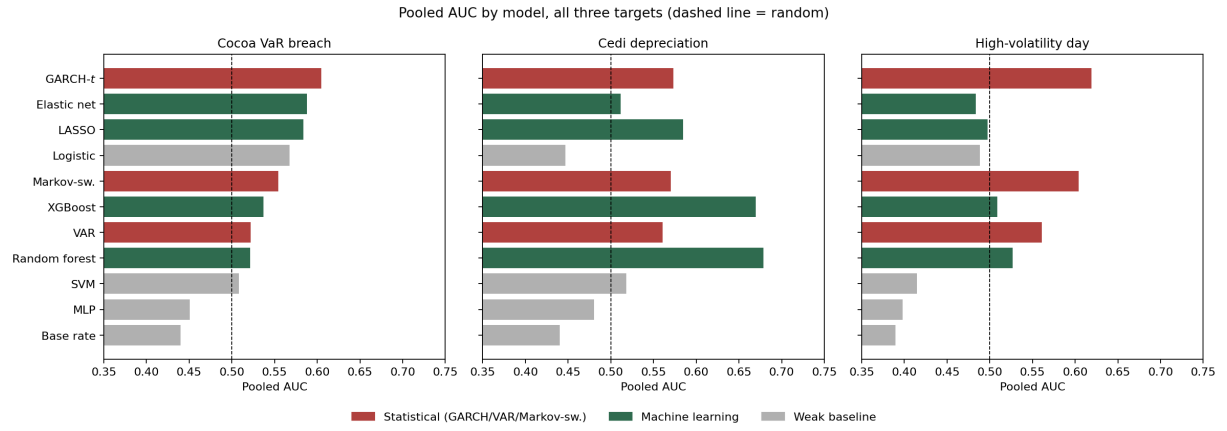


Figure 1: Pooled AUC by model, all three targets. Statistical models (GARCH- t , VAR, Markov-switching) in red, machine-learning models in green, weak baselines in grey. The dashed line marks random performance (AUC = 0.5).

Table 1: Fold-averaged AUC (mean across 5 expanding folds and the 2024 stress holdout), top 3 models per target.

Target	Model	AUC (CV mean)	AUC (2024 stress)
Cocoa VaR breach	Markov-switching	0.601	0.606
	VAR	0.599	0.599
	LASSO	0.578	0.560
Cedi depreciation	Random forest	0.722	0.703
	XGBoost	0.664	0.789
	GARCH- t	0.577	0.698
High-vol day	Markov-switching	0.595	0.748
	VAR	0.562	0.720
	LASSO	0.554	0.418

Table 2: Pooled AUC (all out-of-fold predictions concatenated), top 3 models per target – the authoritative ranking for significance testing.

Target	Model	Pooled AUC
Cocoa VaR breach	GARCH- t	0.605
	Elastic net	0.588
	LASSO	0.584
Cedi depreciation	Random forest	0.679
	XGBoost	0.670
	LASSO	0.584
High-vol day	GARCH- t	0.619
	Markov-switching	0.604
	VAR	0.561

The reversal for cocoa VaR breaches and high-volatility days – from Markov-switching/VAR under fold-averaging to GARCH- t under pooling – is itself informative: it indicates the newer statistical models’ apparent fold-averaged advantage is concentrated in folds with fewer observations and correspondingly higher-variance AUC estimates, exactly where a fold-averaged summary is least trustworthy. Cedi depreciation is robust to the choice: random forest leads under either summary.

5.3. Significance testing

Table 3 summarizes the correction results. Of thirty-one tests (the pooled-AUC leader per target against every other model, plus GARCH- t explicitly where not already the leader), thirteen are nominally significant at 5% uncorrected; four survive full family-wise Bonferroni correction; nine survive the independent Benjamini–Hochberg FDR procedure. Every test surviving either correction involves a serious model beating one of the weakest baseline models (base rate, unregularized logistic regression, the neural network, or the support vector machine); **no test in which both models are genuine, serious contenders survives either correction, for any target.**

Table 3: Significance test summary across all three targets.

Criterion	Tests surviving (of 31)
Uncorrected (5%)	13
Family-wise Bonferroni	4
Benjamini–Hochberg FDR	9

The most robust individual result: random forest significantly beats elastic net, logistic regression, and the base rate for cedi depreciation under *both* corrections – the only comparisons in the entire study to survive the more stringent Bonferroni bar. Random forest does not, however, significantly beat XGBoost, GARCH- t , the vector autoregression, or Markov-switching for the same target at any correction level, meaning the honest summary of the cedi result is "tree ensembles beat weak models decisively" rather than "machine learning beats statistics."

5.4. Interpretability

Permutation-importance and SHAP diagnostics were recomputed on the actual final tuned models from the nested cross-validation run, after an initial pass (run before the model roster and nested tuning were finalized) was identified as stale and discarded rather than reported alongside superseding results.

For **cedi depreciation**, permutation importance and SHAP substantially agree with each other: the cedi’s own short-term behaviour – five-day lagged momentum, one-day lag, five-day momentum – dominates, with Brent momentum and a cocoa–WTI rolling correlation as secondary, genuinely cross-market contributors. Cross-fold stability is modest but positive ($\rho = 0.142$). This corrects, and supersedes, an earlier informal finding (based on the pre-nested-CV pipeline) that identified gold’s own volatility as the dominant predictor; that result does not survive proper tuning and should not be cited.

For **cocoa value-at-risk breaches**, cross-fold stability is statistically indistinguishable from zero ($\rho = -0.009$): the elastic-net model’s top coefficients (Brent and WTI volatility features dominate the fitted coefficients) reshuffle almost completely from fold to fold. This is not evidence of a broken model – predictions are well-behaved and non-degenerate – but evidence that no stable, reproducible feature pattern exists for this target in this feature set, directly corroborating the significance-testing result in Section 5.3 from an independent diagnostic angle.

For **high-volatility days**, the pattern is the same as cocoa ($\rho = -0.030$), consistent with this target’s own weak overall performance (the leading feature-based model, random forest, achieves pooled AUC of only 0.527, barely above chance).

The vector autoregression and Markov-switching model are interpreted on their own native terms rather than forced through a feature-importance framework built for models that consume a feature matrix. The VAR’s cocoa equation shows gold’s first lag (0.052) and the cedi’s first lag (0.043) as its largest cross-market coefficients – modest linear effects, consistent with, not contradicting, the absence of a strong stable pattern documented above. The Markov-switching model identifies genuine two-regime structure in both cocoa and the cedi: cocoa’s volatile regime is short-lived and self-correcting (84% probability of remaining in the quiet regime once there, only 48% of remaining

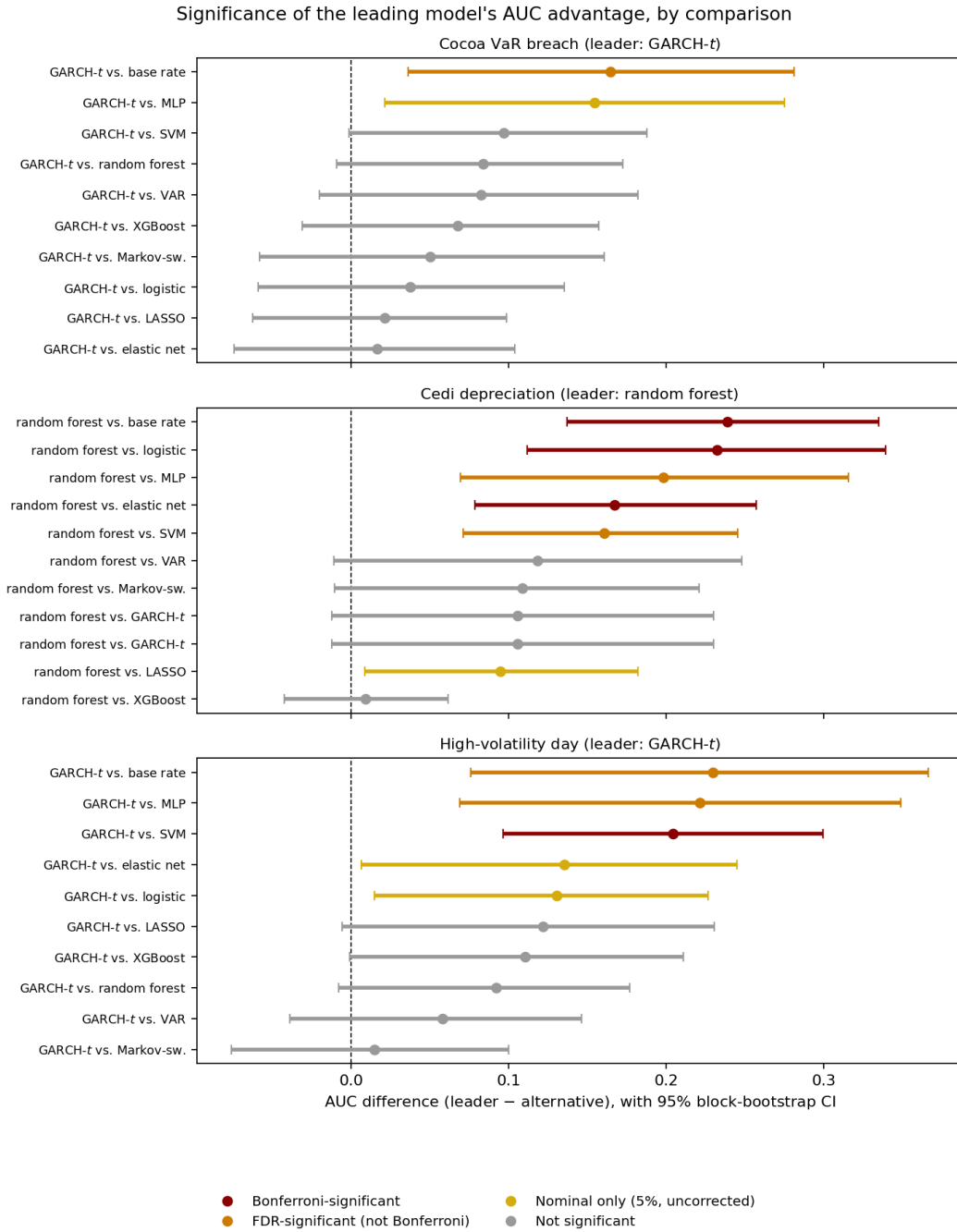


Figure 2: AUC difference (pooled-AUC leader minus each alternative), with 95% block-bootstrap confidence intervals, by target. Marker colour indicates the strictest correction each comparison survives. Every comparison surviving Bonferroni or FDR correction involves a weak baseline model; no comparison between two serious contenders survives either correction.

Cross-fold stability of permutation importance rankings

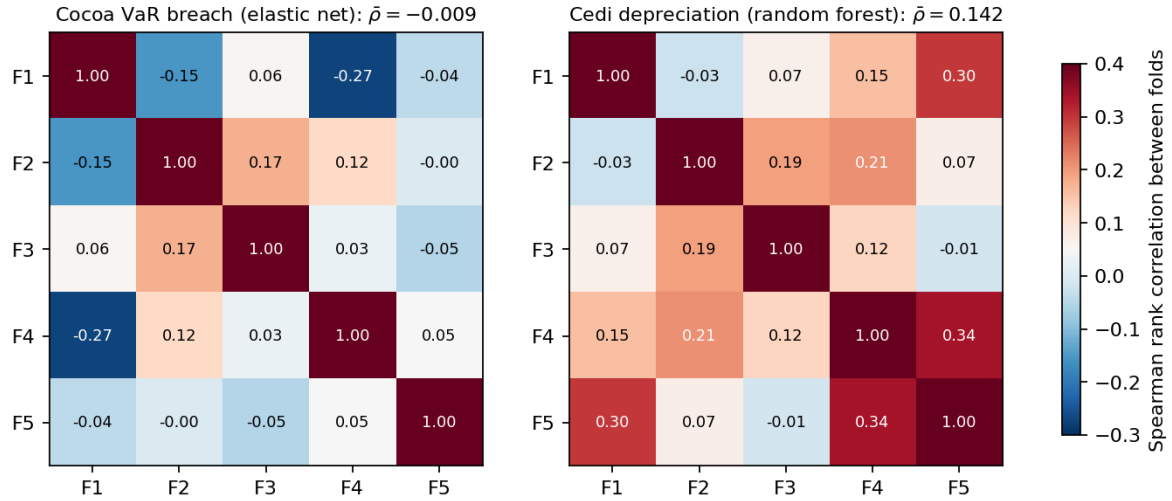


Figure 3: Cross-fold Spearman rank correlation of permutation importance, cocoa VaR breach (elastic net) versus cedi depreciation (random forest). Cocoa's pairwise correlations scatter around zero, mixing positive and negative values across fold pairs; cedi's are modest but consistently positive.

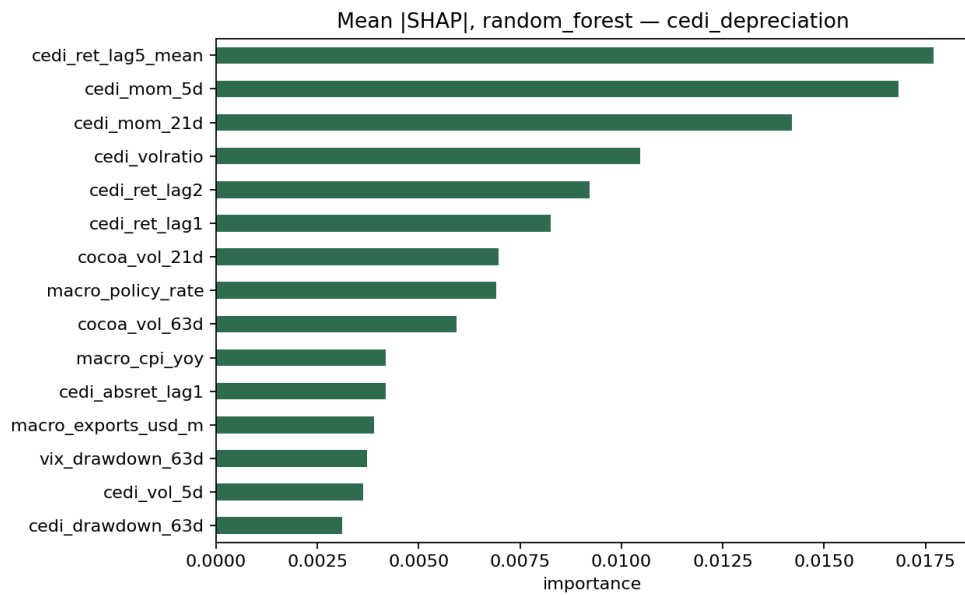


Figure 4: Mean |SHAP| importance, random forest, cedi depreciation, final tuned model trained through the 2024 stress-holdout cutoff. The cedi's own recent behaviour dominates; this supersedes an earlier, pre-nested-CV finding that identified gold volatility as the top predictor, which does not survive proper tuning.

in the volatile regime), while the cedi's regimes are markedly stickier (90% and 65% respectively), consistent with a managed float that holds a stance for extended periods.

6. Discussion

Model risk, not a paradigm winner, is the finding. The disciplined reading of this paper's results is not "machine learning outperforms statistics" or its reverse, but that this data cannot statistically distinguish a best model among the serious contenders for two of three targets, and can only partially distinguish one for the third. This mirrors, deliberately and by design, the companion tail- dependence study's own central finding (Nelson, 2026): that paper found real, substantial commodity-commodity dependence whose *shape* no standard copula family could adequately capture; this paper finds real, substantial predictive gains over naïve baselines whose *specific best model* this sample size cannot pin down. Both results are properly read as findings about the limits of what the available data supports, not failures of the analysis – and in both cases, it is the rigor of the surrounding methodology (goodness-of-fit testing in the companion paper; nested cross-validation and multiple- testing correction here) that makes the null result credible rather than merely an absence of a positive one.

Practical implication for risk-monitoring system design. An institution building a similar tail-risk monitor for a commodity- dependent economy should not treat a single model's cross-validated AUC advantage as sufficient grounds for deployment without the two checks this paper applies: a paired significance test against the leading alternative, and correction for however many such comparisons the monitoring system as a whole entails. The reversal between fold-averaged and pooled rankings in Section 5 is a second, concrete illustration of the same caution: two reasonable summary statistics of the identical underlying predictions can suggest different winners, and reporting only the more favorable one – a temptation this paper resists throughout – would overstate the certainty a monitoring system could actually offer.

Regularization, not model class, may be the more useful lesson for cedi risk specifically. That LASSO and elastic net substantially outperform unregularized logistic regression, while random forest and XGBoost's advantage over the other serious statistical competitors does not survive correction, suggests that for an institution with modest data and modest computational resources, a well-regularized linear model may be a more defensible starting point than either an unregularized classical baseline or a complex ensemble – a more modest, more immediately actionable conclusion than either "use machine learning" or "use statistics."

The cedi's own short-term momentum, not global risk sentiment, appears to be the most reliable available signal. The corrected interpretability finding – the cedi's own recent behaviour dominates, where an earlier, since-superseded analysis had suggested gold volatility – is itself a caution about premature interpretation: economically appealing stories (global risk-off sentiment transmitting through gold to emerging-market currencies) should be treated skeptically until confirmed on properly tuned, stability-checked final models, not provisional ones.

7. Limitations

Sample size, especially for the rarest target. Cocoa VaR breaches occur on only 5.4% of days, giving roughly 150 positive examples across the full eleven-year sample and as few as 55 in the smallest cross-validation fold. This is the direct cause of both the near-zero feature-importance stability documented in Section 5.4 and the inability to statistically distinguish a leading model in Section 5.3, and is a constraint on what any model, however sophisticated, could be expected to achieve on this target.

Neural sequence models were not built, and this choice was sample-size-driven, not a methodological default. Section 4.1 quantifies why: this sample is roughly an order of magnitude below what comparable published sequence-model successes in finance require, and those successes typically rely on cross-sectional pooling unavailable for a single currency. A multi-country pooled extension – across several commodity-dependent economies – is a natural, larger future study this paper's design does not attempt.

Hyperparameter grids were modest, not exhaustive. Two to four values per hyperparameter were searched per model, a deliberate, disclosed scope choice to keep the nested cross-validation's total computational cost proportionate; a more exhaustive search could shift individual point estimates, though is unlikely to overturn the paper's central finding that the serious contenders are not statistically distinguishable from each other, since that finding already accounts for the additional variance a wider search would introduce.

The Bonferroni correction’s scope is the thirty-one tests actually conducted, not every test that could have been conducted. A fully exhaustive pairwise comparison across all eleven models for all three targets would constitute a larger family of tests and a more conservative bar; we test the pooled-AUC leader against every alternative rather than every pairwise combination, a standard and interpretable, but not the only defensible, scope choice.

The VAR is restricted to five core series. VIX and the dollar index, while available as features for the machine-learning models, are excluded from the vector autoregression specifically to keep its parameter count estimable in the smallest cross-validation folds; a larger VAR incorporating these series is a natural extension.

8. Conclusion

We compare eleven statistical and machine-learning models for three daily commodity and currency risk signals in Ghana, using nested time- series cross-validation, a genuine multivariate and regime-switching statistical competitor set (not only GARCH), and paired, multiple-testing-corrected significance tests on pooled out-of-fold predictions – a combination of rigor this paper’s literature review found no directly comparable precedent for, applied specifically to the Ghana cedi. Every serious model clearly outperforms naïve baselines, but after correction, no model significantly outperforms the other serious contenders for cocoa risk or high-volatility days, and only a partial result survives for cedi depreciation. Interpretability diagnostics corroborate this from an independent angle, and a contemporaneous, near-identically designed study of a G7 currency pair reaches a closely parallel conclusion, evidence that this paper’s central finding reflects a genuine property of the model-comparison problem rather than an artifact of this specific emerging-market setting. We read this, as we read the companion tail-dependence study’s own central finding, as a model-risk result: genuine, well-evidenced uncertainty about which sophisticated model is correct, credible precisely because the surrounding methodology – nested tuning, verified fallback behaviour, dual multiple-testing correction, and diagnostics that corroborate rather than merely restate the headline result – is rigorous enough to make that uncertainty a finding rather than an artifact of insufficient testing.

Data availability statement

The complete analysis pipeline, all output tables underlying every figure and table in this paper, and the nested cross-validation and significance-testing code are available at the companion repository referenced in Nelson (2026). Raw commodity futures prices are not redistributed, per Yahoo Finance’s terms of service. Bank of Ghana and FRED source series are redistributed with attribution where their respective terms permit.

Code availability statement

All analysis code (Python 3.12; `scikit-learn`, `xgboost`, `statsmodels`, `arch`, `hmmlearn`, `shap`) is available under an open-source license alongside the companion tail-dependence study’s repository.

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The author declares no conflict of interest.

Author contributions statement

Conceptualization, methodology, software, formal analysis, data curation, writing, and verification: J.N.

Note on the use of AI assistance

Portions of the code implementation, data verification, and manuscript drafting for this paper were carried out with the assistance of an AI language model (Anthropic’s Claude), used as a research and writing tool under the author’s direction, review, and verification throughout. All modelling decisions, empirical results, and their interpretation are the author’s responsibility.

Note on bibliography style

This paper uses the Harvard (author–year) elsarticle bibliography style (`elsarticle-harv.bst`), consistent with the companion tail-dependence study, for readability in a literature-heavy introduction and discussion.

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Appendix A. Feature construction

For each of the seven daily series, the following features are computed at every date t using information available through t only: one- and two-day lagged returns, five-day mean lagged return, rolling volatility at 5/21/63-day horizons, rolling momentum (summed return) at 5/21/63-day horizons, the one-day absolute return, a 63-day drawdown measure, and the ratio of 5-day to 63-day rolling volatility. For cocoa specifically, rolling 21-day correlations with every other series are additionally computed. A joint-tail-count feature records, for each date, the number of the three core commodities (cocoa, gold, Brent) simultaneously below their trailing 10th-percentile return quantile. The lagged macroeconomic block (CPI inflation, exports, policy rate) is merged with a one-month publication lag and forward-filled through publication gaps as described in Section 3.

Appendix B. Full model comparison and significance test tables

The complete nested cross-validation results (all eleven models, all three targets, all six outer folds, with per-fold tuning status and selected hyperparameters) and the complete thirty-one-row significance- test table (with both correction flags) are distributed as `nested_cv_full_results.csv` and `significance_tests_corrected.csv` in the data repository referenced in the Data Availability Statement, rather than reproduced in full here.